

Mathematics A

Unit 2 • Chapter OL-T51 • Expertise Q16+ Classified

Cross-session • chapter-organised candidate

8 classified items • 38 marks • 1 chapter(s)

Source-faithful practice with synchronized solutions

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Contents

Unit 2 • Expertise • Unit 2 • Chapter OL-T51 • Expertise Q16+ • 8 items • 38 marks

Chapter	Items	Marks	Starts
01. OL-T51 Vectors	8	38	4

June 2026 is intentionally deferred; November 2025 remains provisional QP-only in this private candidate.

Vectors

Unit 2 • Expertise • 38 marks • 8 item(s)

June 2025 | 4MA1-1H Q23 | 5 marks

Find an intersection vector from two paths

June 2024 | 4MA1-2H Q24 | 6 marks

Find an intersection vector from two paths

November 2023 | 4MA1-2H Q22 | 5 marks

Recover a division ratio using vector parameters

June 2023 | 4MA1-1H Q19 | 5 marks

Recover a division ratio using vector parameters

November 2024 | 4MA1-1H Q24 | 6 marks

Find an intersection vector from two paths

November 2024 | 4MA1-2H Q17 | 3 marks

Find magnitude of a vector

May-Nov 2020 | 2H Q19 | 3 marks

Use vectors in a parallelogram, trapezium or polygon path

January 2023 | 1H Q24 | 5 marks

Find a section or midpoint vector

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

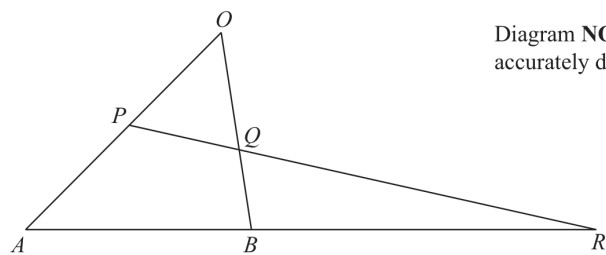


Diagram **NOT**
accurately drawn

OAB is a triangle.
 P is the midpoint of OA
 Q is a point on OB

ABR and PQR are straight lines.

$$\vec{OA} = 12\mathbf{a} \quad \vec{OB} = 8\mathbf{b}$$

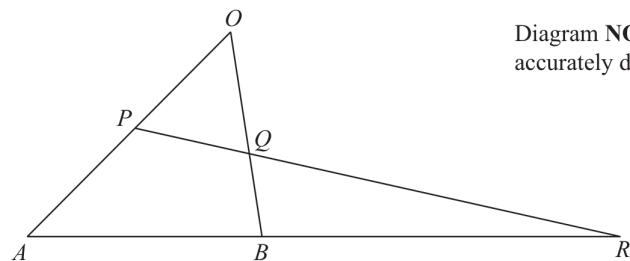
(a) Express $\vec{AB}$ in terms of $\mathbf{a}$ and $\mathbf{b}$

(1)

WORKING SPACE

4MA1-1H Q23

23

Diagram **NOT**
accurately drawn OAB is a triangle. P is the midpoint of OA Q is a point on OB ABR and PQR are straight lines.

$$\vec{OA} = 12\mathbf{a} \quad \vec{OB} = 8\mathbf{b}$$

$$AB:BR = 1:2 \quad \vec{OQ} = n\mathbf{b}$$

(b) Use a vector method to find the value of n

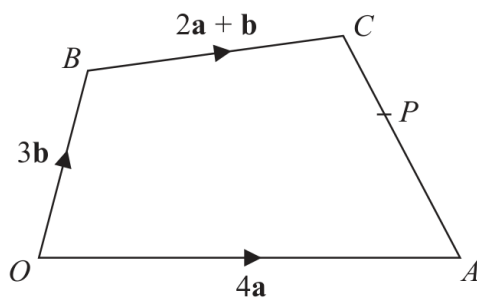
$$n = \dots\dots\dots (4)$$

WORKING SPACE

4MA1-2H Q24

6 marks

24

Diagram **NOT**
accurately drawn

The diagram shows a quadrilateral $OACB$ in which

$$\vec{OA} = 4\mathbf{a} \quad \vec{OB} = 3\mathbf{b} \quad \vec{BC} = 2\mathbf{a} + \mathbf{b}$$

- (a) Find $\vec{AC}$ in terms of $\mathbf{a}$ and $\mathbf{b}$
Give your answer in its simplest form.

$$\vec{AC} = \dots\dots\dots (2)$$

WORKING SPACE

4MA1-2H Q24

6 marks

24

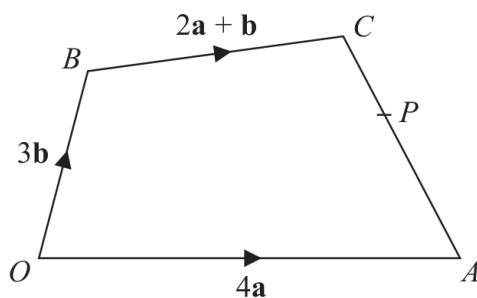


Diagram **NOT**
accurately drawn

The diagram shows a quadrilateral $OACB$ in which

$$\vec{OA} = 4\mathbf{a} \quad \vec{OB} = 3\mathbf{b} \quad \vec{BC} = 2\mathbf{a} + \mathbf{b}$$

The point P lies on AC such that $AP:PC = 3:2$

The point Q is such that OPQ and BCQ are straight lines.

- (b) Using a vector method, find $\vec{OQ}$ in terms of $\mathbf{a}$ and $\mathbf{b}$
Give your answer in its simplest form.
Show your working clearly.

$$\vec{OQ} = \dots\dots\dots (4)$$

Continue your method

6 marks

WORKING SPACE

4MA1-2H Q22 (a)(i), (a)(ii), (b)

5 marks

22

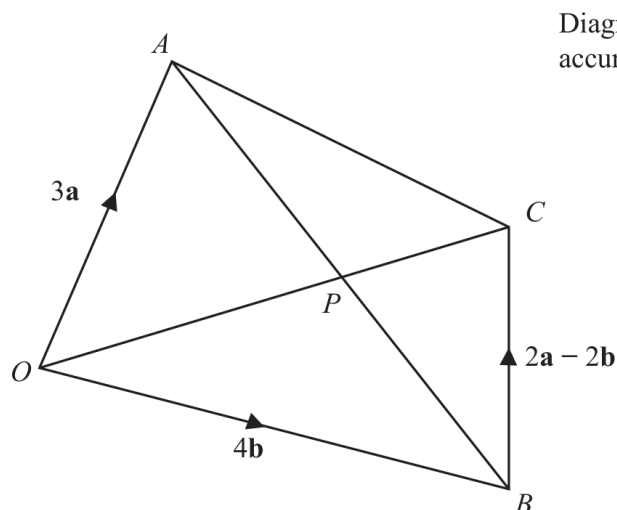


Diagram **NOT**
accurately drawn

$OACB$ is a quadrilateral.

$$\vec{OA} = 3\mathbf{a} \quad \vec{OB} = 4\mathbf{b} \quad \vec{BC} = 2\mathbf{a} - 2\mathbf{b}$$

- (a) (i) Find the vector $\vec{OC}$ in terms of $\mathbf{a}$ and $\mathbf{b}$
Simplify your answer.

$$\vec{OC} = \dots\dots\dots$$

(1)

- (ii) Find the vector $\vec{AB}$ in terms of $\mathbf{a}$ and $\mathbf{b}$

$$\vec{AB} = \dots\dots\dots$$

(1)

The point P lies on AB and on OC

- (b) Using a vector method, find the ratio $AP : PB$
Show your working clearly.

$$\dots\dots\dots$$

(3)

Continue your method

5 marks

WORKING SPACE

4MA1-1H Q19

5 marks

19

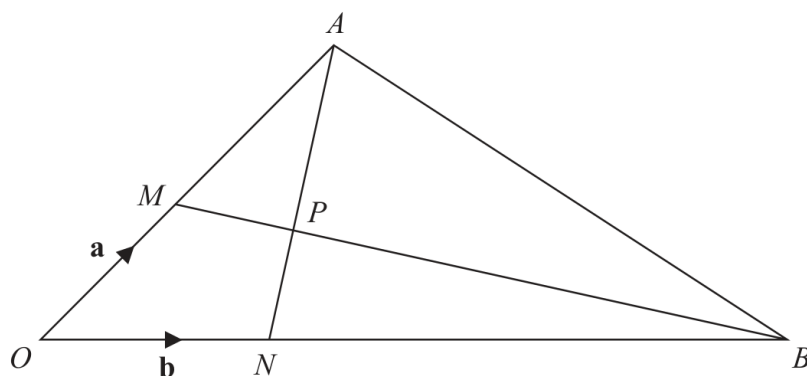


Diagram **NOT**
accurately drawn

OMA , ONB , MPB and NPA are straight lines.

M is the midpoint of OA

$ON:NB = 1:5$

$$\vec{OM} = \mathbf{a} \quad \vec{ON} = \mathbf{b}$$

(a) Find in terms of $\mathbf{a}$ and $\mathbf{b}$ the vector $\vec{AN}$

.....
(1)

(b) Use a vector method to find the ratio $AP:PN$

$AP:PN =$
(4)

Continue your method**5 marks**

WORKING SPACE

4MA1-1H Q24 M01, M02, M03

6 marks

24 OAB is a triangle.

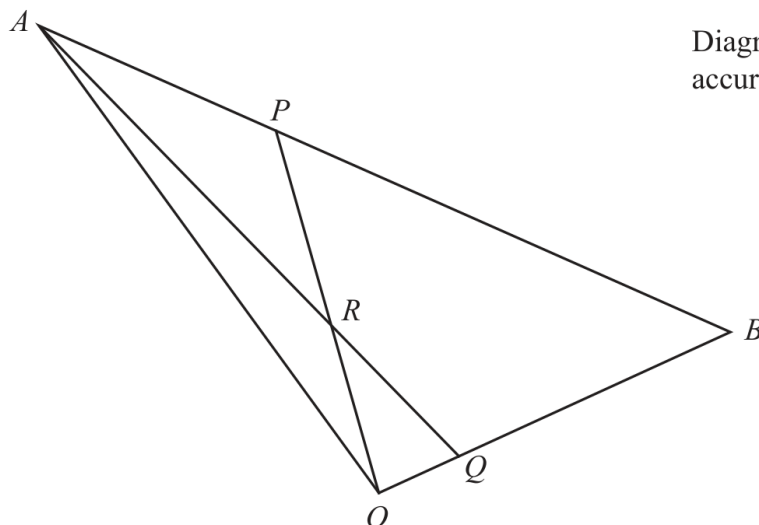


Diagram **NOT**
accurately drawn

$$\vec{OA} = 10\mathbf{a} \quad \vec{OB} = 10\mathbf{b}$$

ARQ and ORP are straight lines.

$$\vec{AP} = \frac{1}{4} \vec{AB} \quad \text{and} \quad \vec{OQ} = \frac{1}{5} \vec{OB}$$

Write the following vectors in terms of $\mathbf{a}$ and $\mathbf{b}$
Simplify your answers.

(i) $\vec{AQ}$

.....
(1)

(ii) $\vec{OP}$

.....
(1)

4MA1-1H Q24 M01, M02, M03

6 marks

(iii) $\vec{OR}$
(4)

WORKING SPACE

4MA1-2H Q17 M01

3 marks

17 Here are two vectors.

$$\vec{FG} = \begin{pmatrix} -5 \\ 2 \end{pmatrix} \quad \vec{HG} = \begin{pmatrix} 4 \\ 14 \end{pmatrix}$$

Calculate the magnitude of the vector $\vec{HF}$

.....

WORKING SPACE

19 OAB is a triangle.

$$\vec{OA} = \mathbf{a} \quad \vec{OB} = \mathbf{b}$$

The point C lies on OA such that $OC : CA = 1 : 2$

The point D lies on OB such that $OD : DB = 1 : 2$

Using a vector method, prove that $ABDC$ is a trapezium.

(Total for Question 19 is 3 marks)

YOUR WORKING

24 $OAED$ is a quadrilateral.

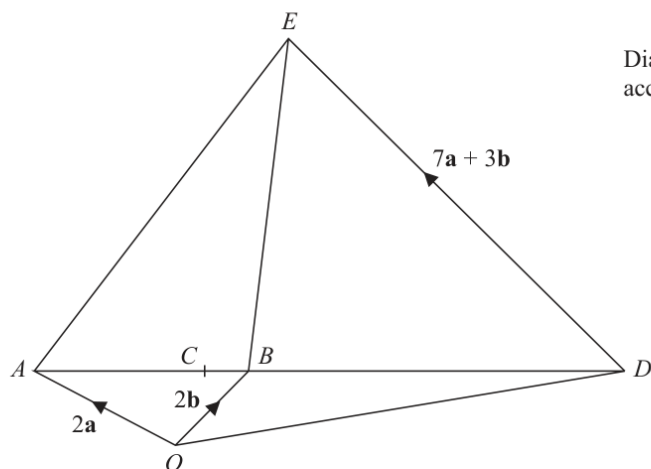


Diagram **NOT**
accurately drawn

$$\vec{OA} = 2\mathbf{a} \quad \vec{OB} = 2\mathbf{b} \quad \vec{DE} = 7\mathbf{a} + 3\mathbf{b}$$

$$AB:BD = 1:2$$

The point C on AB is such that OCE is a straight line.

Use a vector method to find the ratio of $OC:CE$



YOUR WORKING

(Total for Question 24 is 5 marks)

YOUR WORKING